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regsensitivity: Sensitivity Analysis for Omitted Variable Bias in Linear Regression in R

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Abstract

Sensitivity analysis for omitted variable bias has become a routine step in applied regression work, and the methodological literature has produced a sequence of increasingly general results. A recurring limitation of the most widely used tools is that they assume the included control variables are themselves exogenous, an assumption that is rarely defensible and whose failure can reverse a robustness verdict. Diegert, Masten, and Poirier (2026) derive sharp identified sets for a regression coefficient that remain valid when the controls are arbitrarily endogenous, indexed by three interpretable sensitivity parameters. This paper introduces **regsensitivity**, an R package implementing those results together with the related analyses of Oster (2019) and Masten and Poirier (2026), under a single formula-based interface. Beyond porting the methodology, the package contributes bootstrap inference for the breakdown point, which Diegert *et al.* (2026) explicitly leave to future work, and we report a simulation study of its finite-sample coverage. We describe the underlying identification theory, the regime dispatch that the three-parameter problem requires, the package design and API, a replication of the Bazzi, Fiszbein, and Gebresilasse (2020) application that reproduces every value of Diegert *et al.* (2026) Table 1 column (5) and Tables 3 and 4, two further applications, and a comparison with **sensemakr** (Cinelli and Hazlett 2020). A suite of 26 paper-exact regression tests pins the replicated values to within one unit in the last published digit.

Keywords: omitted variable bias, sensitivity analysis, breakdown point, endogenous controls, identified set, R, **regsensitivity**.

1. Introduction

A central concern in empirical research with observational data is that the coefficient of interest in a linear regression may be biased because a relevant covariate has been omitted. The applied literature has long sought tools that quantify this concern: *how much* omitted-variable bias would there have to be in order to overturn the conclusion? Altonji, Elder, and

Taber (2005) introduced the heuristic of comparing the strength of selection on observables to selection on unobservables; Oster (2019) formalized that heuristic and gave a specific quantity δ summarizing the relative strength; Cinelli and Hazlett (2020) recast the question in terms of partial R^2 between the unobservable and the regressors; and Masten and Poirier (2026) identified an asymmetry in the Oster framework that matters when one is interested in sign changes rather than zero crossings. Related approaches include Imbens (2003), Hosman, Hansen, and Holland (2010), Hsu and Small (2013), Krauth (2016) and, in the nonparametric setting, Rosenbaum (2002), Tan (2006) and Masten and Poirier (2018).

Almost all of this work shares an assumption that is easy to overlook: the included control variables are exogenous, in the sense of being uncorrelated with the omitted variable. That assumption is rarely defensible. Controls are typically chosen because they proxy for the same latent forces that the omitted variable represents, so if the omitted variable is a geographic or climatic factor and the controls are geographic and climatic variables, they will be correlated almost by construction. Diegert, Masten, and Poirier (2025) show formally that residualization, the literature's most common device for accommodating endogenous controls, can deliver incorrect robustness conclusions; De Luca, Magnus, and Peracchi (2019), Frölich (2008) and Hünermund and Louw (2025) document further difficulties with control variables in this setting.

Diegert *et al.* (2026), hereafter DMP, respond by deriving sharp identified sets for the coefficient of interest that remain valid when the controls are *arbitrarily* endogenous. Their analysis is indexed by three interpretable sensitivity parameters, and it nests the exogenous-control case as a special case, so the two can be compared within one framework. It is this analysis, together with the closely related work of Oster (2019) and Masten and Poirier (2026), that the **regsensitivity** package implements.

These methods are heavily used—Oster (2019) alone has been cited more than five thousand times—but a complete R implementation has not been available. Section 5 surveys the existing R and Stata ecosystem in detail; in summary, the available R packages implement either Oster's δ alone (**robomit**, Schaub 2025), the Cinelli and Hazlett (2020) robustness value alone (**sensemakr**, Cinelli, Ferwerda, Hazlett, Tsao, Rudkin, and Ljubownikow 2024), or related but distinct diagnostics (**konfound**, Rosenberg, Xu, Lin, Maroulis, Narvaiz, and Frank 2025; **causalsens**, Blackwell 2024; **tipr**, D'Agostino McGowan and Barrett 2024), and none of them allow for endogenous controls or expose the breakdown frontier as a programmable object.

1.1. Contributions

This paper introduces **regsensitivity** for R. The package implements every closed-form regime of the DMP analysis, the nonconvex regime requiring global optimization, the Oster (2019) cubic-root identified set, and the Masten and Poirier (2026) maximum-omitted-variable-bias extension. Concretely, it offers:

- A `formula`-and-`data.frame` interface that fits naturally into existing R workflows, with a control/calibration covariate split expressed through a single `compare` argument.
- Native **ggplot2** (Wickham 2016) rendering of the identified set and the breakdown frontier, with full theming and faceting support.
- Percentile and cluster-robust bootstrap confidence intervals for the breakdown point. DMP state that they "expect the corresponding asymptotic theory for estimation and

inference on the bound functions to be straightforward, but for brevity we do not develop it" (Section 3.6). The package therefore supplies the inferential step that the identification paper leaves open, and Section 10 reports a simulation study of its coverage.

- A reproducibility test layer: 26 **testthat** assertions (across six `test_that` blocks) pin Diegert *et al.* (2026) Table 1 column (5) and Tables 3 and 4 to within one unit in the paper's last printed digit, so future numerical changes—a dependency update, a refactor—break loudly rather than silently.

Section 2 sets out the model and notation. Section 3 presents the DMP identification results and maps each to the function that implements it. Section 4 covers the Oster and Masten–Poirier analyses and gives a crosswalk between the three parameterizations. Section 5 surveys existing software. Section 6 describes the package design. Section 7 is a tutorial introduction to the API. Section 8 replicates the DMP application. Sections 9 and 10 apply the package to a California schools regression and to a simulation study of bootstrap coverage. Section 11 compares with **sensemakr**. Section 12 reports benchmarks and Section 13 concludes.

2. The model

Let Y and X be observed scalar variables, let W_1 be a vector of observed covariates of dimension d_1, and let W_2 be an unobserved vector. Write $W := (W_1, W_2)$. Throughout we assume the variance matrix of (Y, X, W_1, W_2) is finite and positive definite, which is what makes the population linear regressions below well defined. The long regression is

$$Y = \beta_{\text{long}} X + \gamma_1' W_1 + \gamma_2' W_2 + Y^{\perp X,W}, \tag{1}$$

where $Y^{\perp X,W}$ denotes the linear projection residual plus intercept, and is by construction uncorrelated with (X, W). The target of inference is β_{long}. The first-stage regression of treatment on the covariates is

$$X = \pi_1' W_1 + \pi_2' W_2 + X^{\perp W}. \tag{2}$$

Although (2) need not be causal, it is natural to read π_1 as "selection on observables" and π_2 as "selection on unobservables." The baseline assumption of no selection on unobservables is $\pi_2 = 0$, under which β_{long} equals β_{med}, the coefficient on X in the feasible *medium* regression

$$Y = \beta_{\text{med}} X + \gamma_1' W_1 + U_{\text{med}}. \tag{3}$$

Two points about this baseline deserve emphasis, because they shape everything that follows. First, $\pi_2 = 0$ permits W_1 and W_2 to be *arbitrarily* correlated: it restricts the relationship between the covariates and treatment, not the relationship among the covariates. Second, under $\pi_2 = 0$ the coefficient γ_1 on the observed controls is completely unidentified, so control coefficients from the medium regression carry no information even when β_{long} is point identified. Both observations are due to DMP (Proposition 1).

In practice one doubts $\pi_2 = 0$, and the question becomes how far it can fail before a conclusion about β_{long} is lost.

2.1. Control and calibration covariates

Applied specifications usually contain covariates that a researcher would never want to use as a yardstick for the omitted variable—fixed effects, survey-wave indicators, demographic controls. Following DMP (Section 3.4) we therefore split the observed covariates into *control* covariates W_0, which are adjusted for but not used for calibration, and *calibration* covariates W_1, against which the omitted variable's importance is measured. Equations (1)–(2) become

$$Y = \beta_{\text{long}} X + \gamma_0' W_0 + \gamma_1' W_1 + \gamma_2' W_2 + Y^{\perp X,W}, \tag{4}$$

$$X = \pi_0' W_0 + \pi_1' W_1 + \pi_2' W_2 + X^{\perp W}. \tag{5}$$

By the Frisch–Waugh–Lovell theorem, projecting $Y^{\perp W_0}$ on $(1, X^{\perp W_0}, W_1^{\perp W_0}, W_2^{\perp W_0})$ recovers the same coefficients, so every result below applies verbatim after replacing (Y, X, W_1, W_2) with their W_0-residualized counterparts. This is exactly what the package does: the `compare` argument names the members of W_1, and everything else on the right-hand side of the formula becomes W_0 and is partialled out in the first stage of the pipeline described in Section 6.

3. The Diegert–Masten–Poirier analysis

3.1. Measuring selection

DMP measure the strength of selection on unobservables by the standard deviation of the index of unobservables in (2), and compare it to the same quantity for the observables:

$$r_X := \frac{\sqrt{\text{var}(\pi_2' W_2)}}{\sqrt{\text{var}(\pi_1' W_1)}}. \tag{6}$$

Because the indices $\pi_1' W_1$ and $\pi_2' W_2$ are each invariant to invertible linear transformations of W_1 and W_2, so is r_X; it is therefore a unit-free measure of the relative magnitude of selection on unobservables. The baseline $\pi_2 = 0$ is $r_X = 0$. There is also an R^2 representation,

$$r_X^2 = \frac{R^2_{X\sim W_1,W_2} - R^2_{X\sim \pi_1' W_1}}{R^2_{X\sim W_1,W_2} - R^2_{X\sim \pi_2' W_2}}, \tag{7}$$

and, under a linearity assumption on potential treatments, r_X is the ratio of the causal effect on X of a one-standard-deviation increase in the unobserved index to that of the observed index.

By analogy, the relative impact of the unobservables on the *outcome* is

$$r_Y := \frac{\sqrt{\text{var}(\gamma_2' W_2)}}{\sqrt{\text{var}(\gamma_1' W_1)}}, \tag{8}$$

using the coefficients of the long outcome equation (1). The package's three sensitivity parameters are upper bounds on these two ratios plus a bound on control endogeneity:

$$\text{A3:} \quad r_X \le \bar{r}_X, \tag{9}$$

$$\text{A5:} \quad r_Y \le \bar{r}_Y, \tag{10}$$

$$\text{A6:} \quad R_{W_2 \sim W_1 \cdot W_0} \in [\underline{c}, \bar{c}] \subseteq [0, 1], \tag{11}$$

where $R_{W_2 \sim W_1 \cdot W_0}$ is the square root of the R^2 from regressing W_2 on W_1 after partialling out W_0. These correspond to the `rxbar`, `rybar` and `cbar` arguments. The agnostic case $[\underline{c}, \overline{c}] = [0, 1]$ imposes nothing on control endogeneity and is the default. The package fixes $\underline{c} = 0$ and exposes only $\overline{c}$; the general two-sided restriction of DMP Assumption A6 is not currently implemented, a limitation we return to in Section 13.

3.2. Identification from the selection equation alone

Let $\mathcal{B}_I(\overline{r}_X)$ denote the identified set for β_{long} under positive definiteness and A3 only—no restriction whatever on the outcome equation, and no restriction on the correlation between W_1 and W_2. DMP Theorem 1 gives its endpoints in closed form. Writing $\underline{B}$ and $\overline{B}$ for the greatest lower and least upper bounds, if $\overline{r}_X < \sqrt{1 - R^2_{X \sim W_1}}$ then

$$[\underline{B}(\overline{r}_X), \overline{B}(\overline{r}_X)] = [\beta_{\text{med}} - \text{dev}(\overline{r}_X),\ \beta_{\text{med}} + \text{dev}(\overline{r}_X)]\,, \tag{12}$$

where

$$\text{dev}(\overline{r}_X) = \sqrt{\frac{\overline{r}_X^2 R^2_{X \sim W_1}}{1 - R^2_{X \sim W_1} - \overline{r}_X^2 R^2_{X \sim W_1}}} \cdot \sqrt{\frac{\text{var}(Y^{\perp X, W_1})}{\text{var}(X^{\perp W_1})}}, \tag{13}$$

and otherwise the bounds are $-\infty$ and $+\infty$.

Four features of (12) matter in practice. The bounds require the researcher to reason about a single parameter. They are symmetric around β_{med} and collapse to the point β_{med} at $\overline{r}_X = 0$, recovering the baseline. Everything on the right-hand side is a function of the variance matrix of (Y, X, W_1) alone, which is why the entire analysis reduces to a summary of that matrix. And the bounds are finite only for $\overline{r}_X < \sqrt{1 - R^2_{X \sim W_1}}$, a threshold the package computes internally and uses to choose a default grid for `rxbar`.

3.3. The breakdown point

Suppose the medium regression yields $\beta_{\text{med}} \geq 0$ and the concern is that $\beta_{\text{long}} \leq 0$, so that a positive finding is an artifact of selection. Define

$$\overline{r}_X^{\text{bp}} := \sup\{\overline{r}_X \geq 0 : b \geq 0 \text{ for all } b \in \mathcal{B}_I(\overline{r}_X)\}. \tag{14}$$

This is the *breakdown point*: the largest amount of selection on unobservables, as a multiple of selection on observables, consistent with the sign conclusion. DMP Corollary 1 gives it in closed form,

$$\overline{r}_X^{\text{bp}} = \left(\frac{R^2_{Y \sim X \cdot W_1}}{\dfrac{1 - R^2_{X \sim W_1}}{R^2_{X \sim W_1}} + R^2_{Y \sim X \cdot W_1}} \right)^{1/2}. \tag{15}$$

Because the term $(1 - R^2_{X \sim W_1})/R^2_{X \sim W_1}$ is nonnegative, (15) implies $\overline{r}_X^{\text{bp}} < 1$ whenever $R^2_{X \sim W_1} < 1$. This ceiling is specific to the arbitrarily-endogenous-control case: it is a consequence of imposing nothing on $R_{W_2 \sim W_1 \cdot W_0}$, and Section 3.5 shows that restricting control endogeneity lets the breakdown point exceed one. In the application of Section 7 the breakdown point is 1.35 at $\overline{c} = 0$ and 1.19 at $\overline{c} = 0.1$, falling to 0.80 at $\overline{c} = 1$.

Equation (15) also makes the comparative statics transparent. The breakdown point rises with the covariate-adjusted association between outcome and treatment, and *falls* as the first-stage relationship between treatment and the calibration covariates strengthens. The second effect is easy to misread as a defect. It is not: the observed covariates are the yardstick, so when they explain treatment well, a proportionally smaller unobservable suffices to overturn the finding. This is why $\overline{r}_X^{\text{bp}}$ must be read together with the calibration values of Section 3.6 rather than against a universal cutoff.

3.4. Adding restrictions on the outcome equation

Bounding only r_X is conservative, since it allows the omitted variable to have unlimited explanatory power for Y. Imposing A5 as well shrinks the identified set, at the cost of a second parameter. Let $\mathcal{B}_I(\overline{r}_X, \overline{r}_Y)$ be the identified set under A1, A3 and A5, with W_2 a scalar normalized to unit variance. This set has no closed-form description; DMP characterize it in their Appendix C and show (Theorem 2) that the associated *breakdown frontier*

$$\overline{r}_Y^{\text{bf}}(\overline{r}_X, b) := \sup\{\overline{r}_Y \geq 0 : \tilde{b} \geq b \text{ for all } \tilde{b} \in \mathcal{B}_I(\overline{r}_X, \overline{r}_Y)\} \tag{16}$$

solves a smooth optimization problem over a four-dimensional space, whose dimension—importantly for applications with many controls—does not grow with d_1. The set of pairs below the frontier, $RR := \{(\overline{r}_X, \overline{r}_Y) : \overline{r}_Y \leq \overline{r}_Y^{\text{bf}}(\overline{r}_X, b)\}$, is the *robust region*, and its size is itself a measure of robustness (Masten and Poirier 2020).

A convenient scalar summary sets $\overline{r}_X = \overline{r}_Y$, the *common maximal impact* assumption, giving $\overline{r}^{\text{bp}}(b) := \sup\{r \geq 0 : b \geq b \text{ for all } b \in \mathcal{B}_I(r, r)\}$. Since restricting the outcome equation can only shrink the identified set, $\overline{r}_X^{\text{bp}} \leq \overline{r}^{\text{bp}}$ always. In the package this is requested by passing `rybar_expr = function(rx) rx`; Section 8 verifies both quantities against DMP Table 1.

3.5. Restricting control endogeneity

To see why $\overline{c}$ enters at all, note that for scalar W_2 with unit variance the omitted variable bias can be written

$$\beta_{\text{med}} - \beta_{\text{long}} = \frac{\gamma_2 \pi_2 \left(1 - R^2_{W_2 \sim W_1}\right)}{\text{var}(X^{\perp W_1})}. \tag{17}$$

Bounds on r_X and r_Y restrict π_2 and γ_2; the remaining free quantity is $R^2_{W_2 \sim W_1}$, and A6 bounds it directly. DMP Theorem 3 shows that the resulting bounds are again closed-form and symmetric about β_{med},

$$[\underline{B}(\overline{r}_X, \underline{c}, \overline{c}), \overline{B}(\overline{r}_X, \underline{c}, \overline{c})] = [\beta_{\text{med}} - \text{dev}, \beta_{\text{med}} + \text{dev}]\,, \tag{18}$$

where

$$\text{dev} = \sqrt{\frac{\overline{z}_X^2}{\text{var}(X^{\perp W_1}) - \overline{z}_X^2}} \cdot \sqrt{\frac{\text{var}(Y^{\perp X, W_1})}{\text{var}(X^{\perp W_1})}}, \tag{19}$$

where the sensitivity parameters enter only through a scalar function $\overline{z}_X(\overline{r}_X, \underline{c}, \overline{c})$ given in the DMP appendix, and the bounds are finite when $\overline{z}_X^2 < \text{var}(X^{\perp W_1})$. Theorem 1 is the special case $[\underline{c}, \overline{c}] = [0, 1]$. The practical payoff is the trade-off it exposes: $\overline{z}_X$ is finite whenever $\overline{r}_X < \overline{c}^{-1}$, so tightening the assumption on control endogeneity buys tolerance for larger selection on unobservables. DMP Theorem 4 extends the frontier of Section 3.4 to this case.

Equation (17) is also the bridge to Cinelli and Hazlett (2020). DMP show in their Appendix A that

$$|\beta_{\text{med}} - \beta_{\text{long}}| = \sqrt{\frac{R^2_{X\sim W_2\cdot W_1}}{1 - R^2_{X\sim W_2\cdot W_1}}} \cdot \sqrt{R^2_{Y\sim W_2\cdot X,W_1}} \cdot \sqrt{\frac{\text{var}(Y^{\perp X,W_1})}{\text{var}(X^{\perp W_1})}}, \tag{20}$$

a decomposition that also appears in Hosman *et al.* (2010) and Cinelli and Hazlett (2020). The two unknown partial R^2 terms in (20) are precisely the quantities **sensemakr** parameterizes directly, whereas A3 and A5 restrict them indirectly through the ratios (6) and (8). Section 11 returns to this correspondence.

3.6. Calibration

Because $\bar{r}_X$ and $\bar{r}_Y$ are relative parameters, their magnitudes only become interpretable against a reference. DMP propose two point-identified reference sets. The first calibrates $\bar{r}_X$: for each calibration covariate k,

$$\rho_k := \frac{\sqrt{\text{var}(\pi_{1,\text{med},k} W_{1k})}}{\sqrt{\text{var}(\pi'_{1,\text{med},-k} W_{1,-k})}}, \tag{21}$$

where $\pi_{1,\text{med}}$ is the coefficient on W_1 from the OLS regression of X on $(1, W_1)$. Thus ρ_k measures the importance of covariate k relative to all the others, on exactly the scale on which r_X measures the importance of W_2 relative to all of W_1. It is computed by `calibrate_rho()`.

The second calibrates $\bar{c}$: for each k,

$$c_k := R_{W_{1k}\sim W_{1,-k}\cdot W_0}, \tag{22}$$

the square root of the R^2 from regressing W_{1k} on the remaining calibration covariates after partialling out W_0. Large c_k values signal that the calibration covariates are themselves highly inter-correlated, which is evidence *against* the exogenous-controls assumption: if W_2 resembles the components of W_1, one should expect $R_{W_2\sim W_1\cdot W_0}$ to resemble the c_k. One may then take $\bar{c} = \max_k c_k$ as a conservative choice. The package returns c_k^2 from `calibrate_partial_r2()`, matching the scale DMP report in their Table 3.

It bears repeating, with DMP and Hosman *et al.* (2010), that these are "reference points for speculation" about a quantity that is unidentified. Nothing guarantees that r_X is smaller than any ρ_k. Their role is to let data discipline a judgment that remains a judgment.

3.7. When is the long-regression coefficient causal?

Nothing above requires β_{long} to be a causal parameter; the results describe a linear-projection coefficient. Applied users almost always want a causal reading, however, and DMP (Section 4) set out the identification strategies under which one is available. We summarize them because the choice determines what belongs in W_1, and therefore what the sensitivity parameters are measured against.

Under *linear unconfoundedness*—potential outcomes linear in treatment and (W_1, W_2), with treatment mean-independent of potential outcomes given the full covariate set—β_{long} is the average causal effect of X on Y. This is the leading case and the one in which the W_1/W_2

distinction is most natural: W_2 is the confounder one worries about, and W_1 the confounders already adjusted for. DMP also treat nonparametric unconfoundedness, *difference-in-differences*, where X is a post-treatment indicator interacted with group and W_2 collects time-varying unobservables threatening parallel trends, and *instrumental variables*, where the analysis is applied to the reduced form or first stage.

The practical implication is that the package's `compare` argument is a substantive choice, not a technical one. Because $\bar{r}_X$ is a *relative* parameter, moving a variable from W_1 to W_0 changes the yardstick and hence the numerical value of the breakdown point. The guidance from Section 3.6 applies: put in W_1 those covariates whose importance for treatment selection is a credible reference for the omitted variable's importance, and put nuisance adjustments—fixed effects, demographics, survey design variables—in W_0. DMP make exactly this choice in the application we replicate, keeping the geographic and climate variables in W_1 and state fixed effects in W_0.

3.8. Estimation and inference

The identification results depend on the observables only through $\text{var}(Y, X, W_1)$, so sample-analogue estimation simply substitutes a consistent estimator of that matrix; β_{med} is the OLS coefficient from the medium regression. DMP proceed this way and do not develop distribution theory, writing that they "expect the corresponding asymptotic theory . . . to be straightforward, but for brevity we do not develop it" (Section 3.6).

Inference on the breakdown point is nonetheless what an applied reader wants, since a breakdown point of 0.80 estimated with wide uncertainty supports a weaker claim than the same value estimated precisely. Analytic inference is awkward here: the map from the variance matrix to the breakdown point is closed-form in some regimes but the solution to a nonsmooth global optimization in others, so the delta method does not apply uniformly. The package therefore implements a nonparametric bootstrap, in both i.i.d. and cluster-resampled form, and reports a percentile interval. Section 10 studies its finite-sample coverage; we find coverage close to nominal at moderate sample sizes and material under-coverage in small samples, which we report as a caveat rather than a guarantee.

4. Related analyses in the package

4.1. Oster (2019)

Oster (2019) formalizes the Altonji *et al.* (2005) heuristic with a parameter δ measuring the degree of proportional selection, together with an auxiliary parameter R^2_{long}, the R^2 that the infeasible long regression would attain. Given a hypothesized R^2_{long}, the identified value of β_{long} solves a cubic, and the breakdown value of δ—commonly called "Oster's delta"—answers how much proportional selection is needed to move β_{long} to a specified value, usually zero. Applied work often follows Oster's rule of thumb of setting $R^2_{\text{long}} = 1.3 \cdot R^2_{\text{med}}$, and a cutoff of $|\delta| > 1$ as evidence of robustness.

Two cautions apply. First, δ is defined under exogenous controls, which is the assumption DMP relax. Second, as DMP document, the breakdown δ is frequently *miscomputed* in applied work: rather than solving Proposition 3 of Oster (2019), authors set an intermediate

	DMP (2026)	Oster (2019)	Cinelli–Hazlett (2020)
Treatment-side	$\overline{r}_X$: bound on ratio of index SDs, (6)	δ: proportional-selection coefficient	$R^2_{X\sim W_2 \cdot W_1}$ directly
Outcome-side	$\overline{r}_Y$: bound on ratio of index SDs, (8)	R^2_{long}	$R^2_{Y\sim W_2 \cdot X, W_1}$ directly
Control endogeneity	$\overline{c}$: bound on $R_{W_2 \sim W_1 \cdot W_0}$	assumed zero	assumed zero
Scalar summary	$\overline{r}^{\text{bp}}_X$, $\overline{r}^{\text{bp}}$	breakdown $\lvert\delta\rvert$	robustness value RV_q
Natural cutoff	compare to ρ_k, (21)	$\lvert\delta\rvert > 1$	compare RV_q to $R^2_{Y\sim X \cdot W_1}$
Bounded above by 1?	yes for $\overline{r}^{\text{bp}}_X$ at $\overline{c} = 1$	no	yes by construction
Package	**regsensitivity**	**regsensitivity**, **robomit**	**sensemakr**

Table 1: Crosswalk between the sensitivity parameterizations implemented in **regsensitivity** and the closely related Cinelli and Hazlett (2020) framework. All three restrict the same bias decomposition (20); they differ in which functionals they treat as primitive and in whether they permit endogenous controls.

expression to zero and solve, which does not give the breakdown point. DMP report finding this error in several papers published in leading economics journals, including in the application we replicate in Section 8, where the published value of -8.55 should be -23.3. The package implements the correct expression, and Section 8 confirms it reproduces DMP's corrected value.

4.2. Masten and Poirier: bounding the bias magnitude

Masten and Poirier (2026) study the effect of omitted variables on the *sign* of a regression coefficient and observe an asymmetry in the Oster framework that matters for sign conclusions. The package exposes their extension through the `maxovb` argument, an explicit upper bound on $|\beta_{\text{med}} - \beta_{\text{long}}|$, which can be given in absolute terms or relative to $|\beta_{\text{med}}|$. Adding such a bound is useful when a researcher is willing to assert that the omitted variable cannot move the coefficient by more than some amount, independently of any statement about selection ratios.

4.3. A crosswalk between parameterizations

The three analyses ask overlapping questions in different coordinates, which is a persistent source of confusion in applied work. Table 1 maps them. The unifying object is the bias decomposition (20): each framework restricts the two unknown partial R^2 terms in a different way.

The row that most often trips readers is the last but one. A breakdown $\overline{r}^{\text{bp}}_X$ of 0.80 and an Oster $|\delta|$ of 23.3 are not in conflict; they are different normalizations, and only the former is bounded by one under arbitrarily endogenous controls. Reporting a value without naming its parameterization is therefore uninformative, and we recommend against the common practice of comparing a $\overline{r}_X$-type breakdown point to the $|\delta| > 1$ cutoff.

Software	Analysis	Endog. controls	Breakdown point	Frontier plot	Bootstrap inference	**ggplot2** output
R						
regsensitivity	DMP, Oster, MP	✓	✓	✓	✓	✓
sensemakr	Cinelli–Hazlett	–	–	–	–	–
robomit	Oster	–	✓	–	–	✓
konfound	Frank et al.	–	✓	–	–	✓
causalsens	selection bias	–	–	–	–	–
tipr	tipping point	–	✓	–	–	–
Stata						
`psacalc`	Oster	–	✓	–	–	n/a
DMP replication code	DMP	✓	✓	✓	–	n/a

Table 2: Software for omitted-variable-bias sensitivity analysis. "Endog. controls" indicates whether the method remains valid when the included controls are correlated with the omitted variable. "Frontier plot" indicates whether the package can display a breakdown frontier in two sensitivity parameters. The table reflects the analyses the packages document; it is not exhaustive, and this is an actively developing area.

5. Existing software

Sensitivity analysis for omitted variable bias is served by a scattered set of packages, each implementing one branch of the literature. Table 2 summarizes those we are aware of in R and Stata, categorized by the analysis implemented and by whether they expose the features an applied user needs.

In R, **sensemakr** (Cinelli *et al.* 2024) implements Cinelli and Hazlett (2020): partial-R^2 sensitivity contours, the robustness value, and benchmark bounding against observed covariates. It is the most polished tool in this space and the closest in spirit to **regsensitivity**, but it assumes exogenous controls and reduces robustness to a scalar rather than exposing a frontier. **robomit** (Schaub 2025) implements Oster (2019), computing δ and the bias-adjusted coefficient, again under exogenous controls. **konfound** (Rosenberg *et al.* 2025) implements the Frölich (2008)-adjacent tradition of Frank and coauthors, asking what fraction of an estimate would have to be invalidated to overturn an inference, and additionally handles the missing-data analogue. **causalsens** (Blackwell 2024) takes a selection-bias approach for causal effects with a propensity-score flavor, and **tipr** (D'Agostino McGowan and Barrett 2024) performs tipping-point analyses in the epidemiological tradition. None of the five allows the controls to be endogenous, and none returns a breakdown point as a first-class object that can be bootstrapped or plotted against a second sensitivity parameter.

In Stata (StataCorp 2023), Oster distributes `psacalc`, which computes δ and bias-adjusted bounds and is the tool used by most published applications of that method; a Stata implementation of Cinelli and Hazlett (2020) also exists. DMP released Stata code accompanying their paper, which is the direct antecedent of the present package; the **regsensitivity** internals deliberately mirror its regime dispatch, and several internal function names are retained from it so that the two can be compared line by line. Notably, there is no R implementation of the DMP analysis prior to this package, and no implementation in any language that supplies inference on the breakdown point.

6. Package design

6.1. A three-stage pipeline

Every analysis in **regsensitivity** flows through three pure stages.

Stage 1, DGP summary. Given a formula and a data frame, the package projects (Y, X, W_1) onto the orthogonal complement of W_0 and computes the variance matrix of the residualized triple, plus a handful of derived scalars: β_{med}, R^2_{med}, $\text{var}(X^{\perp W_1})$, the threshold $\sqrt{1 - R^2_{X \sim W_1}}$ of Section 3.2, and the basis-change matrix used by the global optimizer when $\overline{r}_Y < \infty$. The result is an object of class `regsen_dgp`. Because Section 3 showed that the identification results depend on the data only through this summary, Stage 1 is the only stage that touches the data at all.

Stage 2, analysis. `regsen_bounds()` and `regsen_breakdown()` apply the chosen analysis to the `regsen_dgp` object plus a grid of sensitivity-parameter values.

Stage 3, presentation. The result is an object of class `regsensitivity` carrying the input grid, the result table, the DGP summary and the parsed hypothesis, with `print`, `summary` and `plot` methods. The plot method dispatches on the sub-command (`"bounds"` versus `"breakdown"`) and on the analysis.

Separating the stages is what makes bootstrapping affordable. Inside a bootstrap loop only Stages 1 and 2 are re-run; the presentation layer is irrelevant per replicate. This yields roughly a two-orders-of-magnitude speed-up over the obvious implementation that calls the top-level function on each replicate, and is why the cluster bootstrap in Section 7.8 completes in about a second.

6.2. Regime dispatch

The three-parameter problem is not uniformly tractable, so the package dispatches on which regime the requested triple $(\overline{r}_X, \overline{r}_Y, \overline{c})$ falls into. The internal function responsible is `dmp_identified_set()`. There are four regimes:

1. *$\overline{r}_X$ and $\overline{r}_Y$ both above the finiteness threshold.* The identified set is $(-\infty, +\infty)$ and is returned without computation.

2. *$\overline{r}_Y = \infty$.* Theorem 1 or Theorem 3 applies and the bounds are evaluated in closed form from (13) or (18). This is the fast path, and it covers the default call.

3. *$\overline{r}_Y < \infty$, $\overline{c} = 0$.* The constraint set reduces to a quadratic inequality in a single variable, solved exactly.

4. *$\overline{r}_Y < \infty$, $\overline{c} > 0$.* The bound is the optimum of a nonconvex objective over the cube $[0, 1]^3$, evaluated with the locally-biased DIRECT-L algorithm (Jones, Perttunen, and Stuckman 1993; Gablonsky and Kelley 2001) as implemented in **nloptr** (Ypma and Johnson 2025; Johnson 2024).

Regime 4 is two orders of magnitude slower than the others, as Section 12 quantifies. We verified its convergence directly: at $(\overline{r}_X, \overline{r}_Y, \overline{c}) = (1.1, 1.1, 0.7)$ on the application data the returned lower bound is stable to four decimal places as the evaluation budget increases from

10^3 to 10^5 function calls, so the default budget of 200 evaluations per grid point is not the accuracy-limiting factor.

There is one region DMP do not characterize, namely $\overline{r}_X > r_{\max}(\overline{c}) > \overline{r}_Y$, where $r_{\max}(\overline{c})$ is the finiteness threshold. Rather than return a possibly wrong number the package raises an error naming the region. Users encountering it should either lower $\overline{r}_X$ below $r_{\max}(\overline{c})$ or raise $\overline{r}_Y$ above it; Section 7.9 shows the practical consequence when tracing a frontier.

6.3. Object structure

The package defines three classes. `regsen_dgp` is the Stage 1 summary: it holds the residualized variance matrix and the derived scalars β_{med}, R^2_{med}, $\text{var}(X)$, $\text{var}(X^{\perp W_1})$ and the basis-change matrix. It is deliberately small, since it is what the bootstrap recomputes R times.

`regsensitivity` is the user-facing result: a list carrying the sub-command, the analysis label, the call, the sample size, the variable names, the sensitivity-parameter grid, the summary statistics, the `regsen_dgp` object and a `results` data frame with one row per grid point. For a bounds analysis `results` has columns `rxbar`, `rybar`, `cbar`, `bmin` and `bmax`; for a breakdown analysis it has `index` and `breakdown`. These are ordinary data frames, so downstream manipulation needs no accessor functions, as Section 7 illustrates throughout.

`regsensitivity_boot` holds the bootstrap output: the point estimate, the vector of replicates, the percentile interval, the level, R, the cluster variable and a count of failed replicates. Retaining the raw replicates lets users compute alternative intervals or inspect the bootstrap distribution directly, which we recommend given the coverage results of Section 10.

6.4. Numerical robustness

Two classes of numerical hazard arise. First, quantities that are mathematically guaranteed to lie in $[0, 1]$—ratios of residual to total variance, for instance—can exceed one by rounding, producing negative arguments to square roots. Each such site clamps its argument. Second, the DIRECT-L search explores parameter points at which the expansion of the constraint set is degenerate, yielding `NA` or `NaN` in intermediate coefficients. Every comparison in the quadratic-inequality solver is therefore wrapped so that a missing value cannot propagate into an `if` condition, and the objective returns $+\infty$ at infeasible points so the optimizer moves away from them. An earlier version of the package produced platform-dependent results on Windows for exactly this reason: the optimizer visited slightly different points there and hit these edge cases more often.

7. The package API

We illustrate the package by replicating the Diegert *et al.* (2026) application to the Bazzi *et al.* (2020) *Frontier Culture* study, which asks whether longer historical exposure to the American frontier produced durable "rugged individualism." The data are bundled with the package. The specification below is column (5) of DMP Table 1: the outcome is the county-level average Republican presidential vote share over 2000–2016, the treatment is total frontier experience, the calibration covariates W_1 are ten geographic and climate variables, and W_0 contains state fixed effects.

```
R> data("bfg2020", package = "regsensitivity")
R> bfg2020$statea <- factor(bfg2020$statea)
R> w1 <- c("log_area_2010", "lat", "lon", "temp_mean", "rain_mean",
+          "elev_mean", "d_coa", "d_riv", "d_lak", "ave_gyi")
R> form <- reformulate(c("tye_tfe890_500kNI_100_l6", w1, "statea"),
+                      response = "avgrep2000to2016")
```

The formula lists the treatment first; every other right-hand-side term is a control. The `compare` argument then names which controls belong to W_1, with the remainder becoming W_0. Here `compare = w1` puts the ten geographic variables in W_1 and the state factor in W_0, implementing the split of Section 2.1.

7.1. Bounds across the selection parameter

`regsen_bounds()` is the main entry point. Called with a scalar `cbar` and no `rxbar`, it sweeps `rxbar` across $[0, r_{\max}(\bar{c})]$, the range over which the identified set is finite.

```
R> bnds <- regsen_bounds(form, bfg2020, compare = w1, cbar = 0.1)
R> print(bnds)


Regression Sensitivity Analysis ----- Bounds
-------------------------------------------------------------------------
Analysis:          DMP (2026)
Treatment:         tye_tfe890_500kNI_100_l6
Outcome:           avgrep2000to2016
N (obs):           2036
Hypothesis:        Beta > 0
Breakdown point:   1.1947

--- Summary statistics ----------------------------------
  Beta (short)                 1.9246
  Beta (medium)                2.0548
  R2 (short)                   0.0328
  R2 (medium)                  0.1051
  Var(Y)                     101.7387
  Var(X)                       0.9014
  Var(X_Residual)              0.8823

--- Results ---------------------------------------------
   rxbar rybar   cbar       bmin   bmax
       0  +Inf    0.1     2.0548 2.0548
 0.40807  +Inf    0.1     1.4221 2.6875
 0.81613  +Inf    0.1    0.72444 3.3851
  1.2242  +Inf    0.1 -0.060227 4.1697
  1.6323  +Inf    0.1   -0.96596 5.0755
  2.0403  +Inf    0.1    -2.0488 6.1583
```

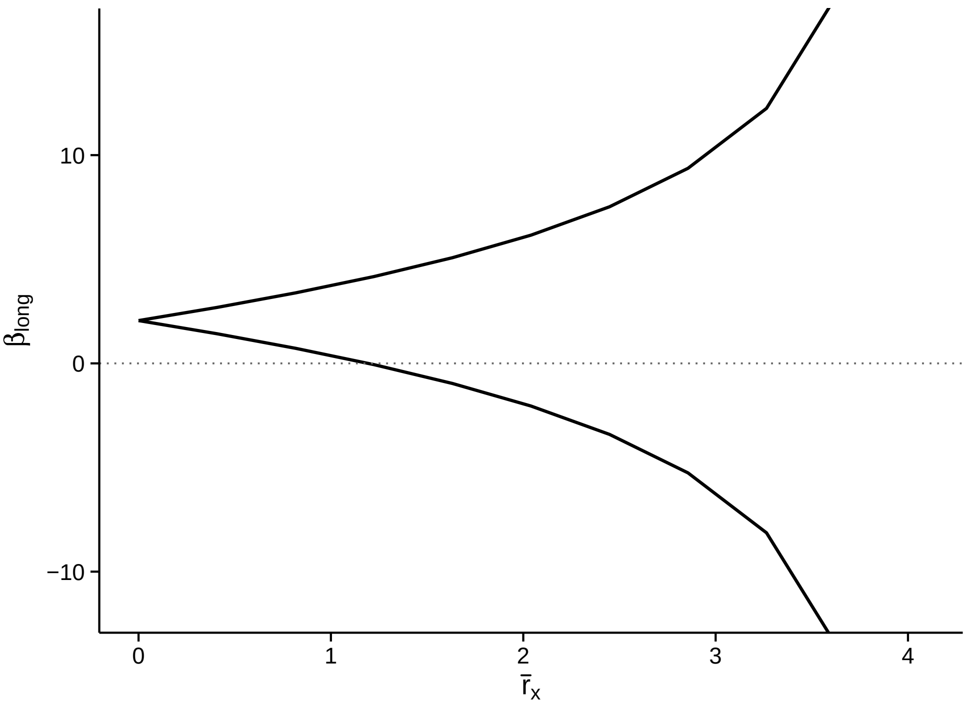

Figure 1: Estimated identified set for the long-regression coefficient as a function of rxbar, holding cbar at 0.1. The two curves are the lower and upper bounds of the identified set; the band between them widens continuously from the point beta-med at rxbar = 0 and diverges at the finiteness threshold. The horizontal line marks the hypothesis value zero, and the breakdown point is where the lower bound crosses it.

```
 2.4484  +Inf    0.1   -3.4099 7.5195
 2.8565  +Inf    0.1   -5.2589 9.3684
 3.2645  +Inf    0.1   -8.1355 12.245
 3.6726  +Inf    0.1   -14.208 18.317
 4.0807  +Inf    0.1      -Inf   +Inf
```

The header reports the estimation sample ($n = 2036$ counties), the parsed hypothesis, the breakdown point, and the summary statistics that Section 6 identified as the only data-dependent inputs. Note $\widehat{\beta}_{\text{med}} = 2.0548$, which is the value DMP Table 1 column (5) reports as 2.055. The breakdown point at $\bar{c} = 0.1$ is 1.1947: the smallest $\bar{r}_X$ at which the hypothesis $\beta_{\text{long}} > 0$ first fails. Consistent with Section 3.3, it exceeds one here because $\bar{c} = 0.1$ restricts control endogeneity; at $\bar{c} = 1$ it falls to 0.8036 and at $\bar{c} = 0$ it rises to 1.3500.

The `results` table is an ordinary data frame, so the identified set can be manipulated directly, and the `plot` method renders it with **ggplot2**.

```
R> plot(bnds)
```

Because the return value is a **ggplot2** object, the usual grammar applies: adding a theme, new labels, a facet or a scale all work without special support from the package.

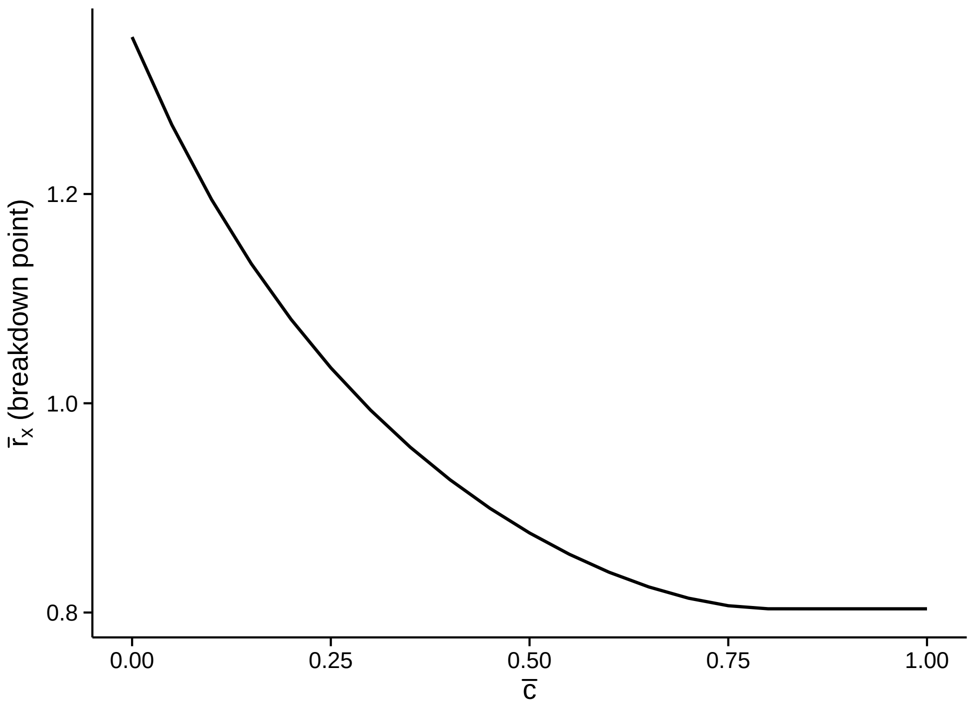

Figure 2: Estimated breakdown point for the hypothesis beta-long > 0 as a function of the bound cbar on control endogeneity. Moving right relaxes the exogenous-controls assumption. The curve is monotonically decreasing and crosses one, so the robustness verdict depends materially on how endogenous the controls are assumed to be.

7.2. Breakdown points across control endogeneity

`regsen_breakdown()` computes the breakdown point rather than the whole identified set, and accepts a vector for the parameter to trace. Sweeping `cbar` shows how much the robustness verdict depends on the exogenous-controls assumption, which is the central question DMP pose.

```
R> bd <- regsen_breakdown(form, bfg2020, compare = w1,
+                         cbar = seq(0, 1, 0.05))
R> head(bd$results, 4)

  index breakdown
1  0.00     1.350
2  0.05     1.266
3  0.10     1.195
4  0.15     1.133

R> plot(bd)
```

Figure 2 makes the paper's motivating point concrete. Under exogenous controls the conclusion survives selection on unobservables up to 135.0% of selection on observables; under arbitrarily endogenous controls only up to 80.4%. A researcher who reported only the former would overstate robustness by a factor of about 1.7.

7.3. Hypotheses other than a sign change

By default the package tests the sign of β_{med} against zero. The `beta` argument, together with the helpers `bnd_lb()`, `bnd_ub()` and `bnd_eq()`, specifies other hypotheses: `bnd_lb(1)` for $\beta_{\text{long}} > 1$, `bnd_eq(0)` for a two-sided statement. Passing a vector traces the breakdown point as a function of the hypothesized value.

```
R> regsen_breakdown(form, bfg2020, compare = w1, cbar = 1,
+                    beta = bnd_lb(c(0, 0.5, 1, 1.5)))$results
```

```
  index breakdown
1   0.0    0.8036
2   0.5    0.7175
3   1.0    0.5753
4   1.5    0.3481
```

The breakdown point falls as the hypothesis becomes more demanding, which it must: concluding $\beta_{\text{long}} > 1.5$ tolerates less bias than concluding $\beta_{\text{long}} > 0$.

7.4. The Oster (2019) analysis

Switching analyses requires only the `analysis` argument. Oster's rule-of-thumb auxiliary parameter is $R^2_{\text{long}} = 1.3R^2_{\text{med}}$, obtained with `r2long_type = "relative"`.

```
R> o <- regsen_breakdown(form, bfg2020, compare = w1,
+                        analysis = "oster",
+                        r2long = 1.3, r2long_type = "relative",
+                        beta = bnd_eq(0))
R> o$results$breakdown
```

```
[1] -23.29
```

This is the corrected Oster delta for this specification. DMP report -23.3 where Bazzi *et al.* (2020) published -8.55; the discrepancy is the miscomputation described in Section 4. Note that the package returns the signed value and that $|\delta|$ is the quantity conventionally compared to one.

7.5. Bounding the bias magnitude directly

The `maxovb` argument implements the Masten and Poirier (2026) extension described in Section 4: an explicit ceiling on $|\beta_{\text{med}} - \beta_{\text{long}}|$, expressed either in the units of the outcome (`maxovb_type = "bound"`) or as a fraction of $|\beta_{\text{med}}|$ (`maxovb_type = "relative"`). It is useful when a researcher can defend a statement about how far the coefficient could move without being willing to commit to a selection ratio.

Oster's equality analysis returns the roots of a cubic, so the identified set can have up to three elements at each δ. Adding a bias ceiling discards the roots that violate it.

```
R> regsen_bounds(form, bfg2020, compare = w1, analysis = "oster",
+                delta = c(0.5, 1, 2), r2long = 1.3,
+                r2long_type = "relative",
+                maxovb = 0.5, maxovb_type = "relative")$results
```

```
  delta r2long beta1 beta2 beta3
1   0.5 0.1367 2.084    NA    NA
2   1.0 0.1367 2.113    NA    NA
3   2.0 0.1367 2.173    NA    NA
```

Here the constraint is that the omitted variable cannot move the estimate by more than half of $|\hat{\beta}_{\text{med}}|$. Root columns that are NA have been excluded by that constraint rather than failing to exist.

7.6. The default summary

`regsen_summary()` reproduces the sweep that the Stata command performs when no subcommand is given: a DMP bounds analysis at the default grid together with an Oster breakdown analysis over a range of R^2_{long} values from the rule-of-thumb value up to one. It returns a list of two `regsensitivity` objects and is the quickest way to see both analyses at once.

```
R> smry <- regsen_summary(form, bfg2020, compare = w1)
R> names(smry)
```

```
[1] "dmp_bounds"      "oster_breakdown"
```

```
R> head(smry$oster_breakdown$results, 4)
```

```
   index breakdown
1 0.1367    0.9739
2 0.2367    0.9732
3 0.3367    0.9723
4 0.4367    0.9714
```

The Oster breakdown is nearly flat in R^2_{long} for this specification, which is worth checking in any application: when it is steep, the reported δ depends heavily on an auxiliary parameter that is rarely defended.

7.7. Calibration helpers

The reference values of Section 3.6 are computed by two functions. `calibrate_rho()` returns the ρ_k of (21), on a percentage scale, sorted by importance.

```
R> rho <- calibrate_rho(form, bfg2020, compare = w1)
R> rho
```

```
        variable    rho
10       ave_gyi 118.34
7          d_coa  78.58
3            lon  49.93
4      temp_mean  37.26
5      rain_mean  29.29
2            lat  26.93
8          d_riv  24.96
1  log_area_2010  22.46
6      elev_mean  20.25
9          d_lak  12.06
```

These are the values of DMP Table 4 column (5), which we verify to published precision in Section 8. Their interpretation is the point of the exercise: the breakdown point at $\bar{c} = 1$ is 80.4%, which exceeds 9 of the 10 calibration values. Only `ave_gyi` is more important for treatment selection than the unobservable would have to be. If one regards the ρ_k as plausible magnitudes for r_X, the finding is fairly robust.

`calibrate_partial_r2()` returns c_k^2 from (22), the diagnostic for whether exogenous controls is tenable at all.

```
R> calibrate_partial_r2(form, bfg2020, compare = w1)
```

```
        variable      R2
4      temp_mean 0.89382
2            lat 0.87613
6      elev_mean 0.68090
10       ave_gyi 0.64751
5      rain_mean 0.55878
7          d_coa 0.48721
3            lon 0.43437
8          d_riv 0.13463
9          d_lak 0.09986
1  log_area_2010 0.09813
```

Several of these are large. Under exogenous controls, W_2 would have a value of zero in this table. Since the omitted variables of concern in this application are themselves geographic and climatic, expecting $R_{W_2 \sim W_1 \cdot W_0}$ to be near zero while the observed covariates reach 0.89 is difficult to defend—which is precisely why the $\bar{c}$-indexed analysis is needed here.

7.8. Bootstrap inference

The breakdown point is a statistic, and `regsen_boot()` attaches a percentile confidence interval to it. Because the Bazzi *et al.* (2020) design is spatial, we resample whole grid cells rather than counties.

```
R> set.seed(2026)
R> boot <- regsen_boot(form, bfg2020,
```

```
+                       compare = w1, cbar = 1,
+                       cluster = "km_grid_cel_code",
+                       R = 199, show_progress = FALSE)
R> print(boot)

Bootstrap confidence interval for the breakdown point
------------------------------------------------------------
  R                  : 199
  Cluster bootstrap  : km_grid_cel_code
  Confidence level   : 95%
  Point estimate     : 0.8036
  95% CI            : [0.5039, 0.8568]
```

The point estimate is the value DMP report as 80.4%. The interval is informative for the substantive question: its lower endpoint, 0.504, still exceeds 8 of the 10 ρ_k values above, so the qualitative conclusion does not rest on the point estimate alone. (The count is computed from the interval, not asserted: the nearest ρ_k sits close to the endpoint, and a different seed could move it across.) Omitting `cluster` gives the i.i.d. bootstrap, which is faster but understates uncertainty when the data are spatially correlated.

7.9. Tracing a breakdown frontier

The frontier (16) trades $\overline{r}_X$ against $\overline{r}_Y$. The package does not expose it as a single call, but it is a short loop over `regsen_breakdown()`, and building it this way illustrates how the primitives compose. For each value of $\overline{r}_Y$ we ask for the $\overline{r}_X$ breakdown point, at three levels of control endogeneity.

```
R> ry_grid <- c(0.5, 0.75, 1, 1.25, 1.5, 2, 3, Inf)
R> frontier <- do.call(rbind, lapply(c(0, 0.5, 1), function(cb) {
+      rx <- vapply(ry_grid, function(ry) {
+          out <- try(regsen_breakdown(form, bfg2020, compare = w1,
+                                      cbar = cb, rybar = ry),
+                     silent = TRUE)
+          if (inherits(out, "try-error")) NA_real_
+          else out$results$breakdown[1]
+      }, numeric(1))
+      data.frame(cbar = cb, rybar = ry_grid, rxbar_bp = rx)
+  }))
R> subset(frontier, cbar == 1)

   cbar rybar rxbar_bp
17    1  0.50   0.9894
18    1  0.75   0.9894
19    1  1.00   0.8917
20    1  1.25   0.8039
21    1  1.50   0.8036
22    1  2.00   0.8036
```

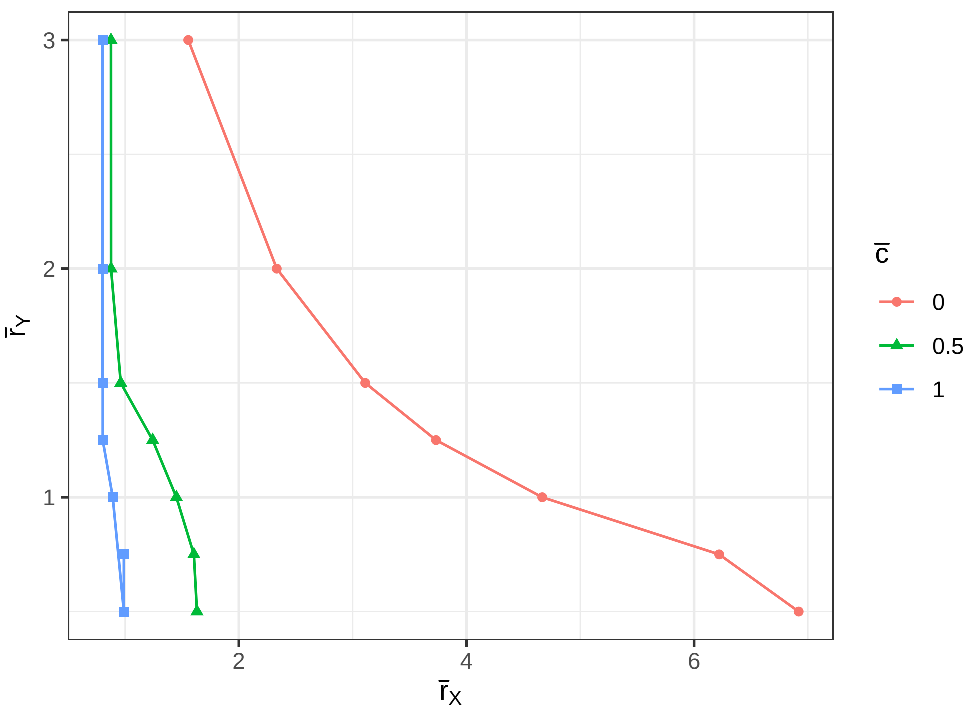

Figure 3: Estimated breakdown frontiers for the hypothesis beta-long > 0. Each curve fixes a bound cbar on control endogeneity and traces the largest rxbar consistent with the conclusion, as a function of rybar. Pairs below and left of a curve are robust. Tightening cbar shifts the frontier outward, so assumptions about control endogeneity and about selection strength are substitutes.

```
23     1  3.00   0.8036
24     1   Inf   0.8036
```

Two things are visible. The frontier moves outward as $\bar{c}$ falls, quantifying the substitution between the two assumptions noted after (18). And the $\bar{c} = 1$ curve is confined to $\bar{r}_X < 0.99$, the finiteness threshold at that $\bar{c}$. The `try()` guard in the loop is defensive: on this grid no call fails, because `regsen_breakdown()` solves along the boundary of the characterized region, but `regsen_bounds()` does raise an error for points beyond it – the region Section 6.2 described as uncharacterized – and the guard costs nothing on grids that stray there. Users who do hit that error should not mistake it for a convergence failure.

8. Replicating Diegert, Masten and Poirier (2026)

The package ships a test suite that pins its output to the published values of the paper it implements. This section reports the comparison; the assertions themselves live in `tests/testthat/test-paper-table1.R` and run on every check.

Table 3 collects the results. All twenty-six assertions pass. Four values differ from the published figure in the last printed digit, which we report explicitly rather than rounding away. Three are in DMP Tables 3 and 4: our $\hat{c}_k^2$ for average temperature is 0.8938 against

Quantity	Published	Package	Source
$\hat{\beta}_{\text{med}}$	2.055	2.055	Table 1, Panel A
Number of counties	2,036	2,036	Table 1, Panel A
Mean of dependent variable	60.04	60.04	Table 1, Panel A
$\hat{\delta}^{\text{bp}}_{\text{resid}}$ (corrected)	-23.3	-23.3	Table 1, Panel B
$\hat{\bar{r}}^{\text{bp}}_X$ (%)	80.4	80.4	Table 1, Panel C
$\hat{\bar{r}}^{\text{bp}}$ (%)	95.9	95.8	Table 1, Panel C

Table 3: Replication of Diegert *et al.* (2026) Table 1, column (5). Published values are as printed in that paper; package values are computed by the code in this section. Tables 3 and 4 of that paper are replicated entry by entry in Sections 7.7 and 8.

a published 0.893, for average rainfall 0.5588 against 0.560, and our $\hat{\rho}_k$ for elevation is 20.25 against 20.2. The fourth is in Table 3 itself, where our $\hat{\bar{r}}^{\text{bp}}$ of 95.8% is printed as 95.9% by DMP. Every entry agrees to within 0.07, and the pattern is consistent with rounding in the published tables rather than with a difference in estimand. We flag them because a reader checking the implementation against the paper would otherwise wonder.

Two of these deserve comment. The corrected Oster delta of -23.3 reproduces DMP's correction of the published -8.55, which is a useful check that the package implements Proposition 3 of Oster (2019) rather than the shortcut that produced the original number. And the ordering $\hat{\bar{r}}^{\text{bp}}_X = 80.4\% \leq \hat{\bar{r}}^{\text{bp}} = 95.8\%$ is the inequality guaranteed at the end of Section 3.4, obtained here through two entirely different code paths—the closed form (15) and a bisection over DIRECT-L solves—which makes it a genuine consistency check rather than a tautology.

9. Application: California test scores

The Stock and Watson (2011) treatment of the `CASchools` data, distributed with **AER** (Kleiber and Zeileis 2008), is a canonical applied regression and a useful second illustration because the substantive question—whether smaller classes raise test scores—has an obvious omitted-variable concern. We regress the average of reading and mathematics scores on the student–teacher ratio, using the English-learner share and the free-lunch share as calibration covariates and county fixed effects as controls.

```
R> suppressPackageStartupMessages(library("AER"))
R> data("CASchools", package = "AER")
R> CASchools$score <- with(CASchools, (read + math) / 2)
R> CASchools$STR   <- CASchools$students / CASchools$teachers
R>
R> f <- score ~ STR + english + lunch + county
R> res <- regsen_bounds(f, CASchools, compare = c("english", "lunch"),
+                       cbar = 1)
R> res$dgp$beta_med

[1] -0.9011
```

```
R> res$breakdown

[1] 0.9446
```

The medium regression gives $\hat{\beta}_{\text{med}} = -0.901$: a one-unit increase in the student–teacher ratio is associated with a 0.90-point lower average score. The breakdown point at $\bar{c} = 1$ is 0.945, so the negative-effect conclusion survives unless selection on unobservables is at least 94.5% as strong as selection on the two calibration covariates. Under exogenous controls the figure is 2.878.

This is a case where calibration changes the reading. The two ρ_k values are 115.9% and 86.3%, so the breakdown point at $\bar{c} = 1$ lies below the larger of them. An unobservable no more important for class-size selection than the free-lunch share is already enough to overturn the sign. The bootstrap interval, $[0.531, 0.997]$ over $R = 199$ i.i.d. replicates, does not change that assessment. Compared with the frontier application of Section 7—where the breakdown point exceeded nine of ten calibration values—this specification is considerably less robust, and the contrast illustrates why a breakdown point should never be read against a universal cutoff.

10. Simulation: coverage of the bootstrap interval

Bootstrap inference on the breakdown point is the package's main addition to the methodology, so it needs evidence rather than assertion. This section reports a simulation study of its finite-sample coverage. We use a design in which the omitted variable is correlated with a calibration covariate, so that control endogeneity is present by construction:

$$\begin{aligned} W_{1a}, W_{1b} \sim N(0,1), \qquad W_2 &= \rho W_{1a} + \sqrt{1-\rho^2}\,\varepsilon, \\ X &= 0.5W_{1a} + 0.3W_{1b} + 0.5W_2 + \nu, \\ Y &= 1.0X + 0.3W_{1a} + 0.2W_{1b} + 0.4W_2 + u, \end{aligned}$$

with $\rho = 0.5$ and all disturbances standard normal. Because the breakdown point is a functional of the population variance matrix rather than a structural parameter, there is no closed-form "true" value to compare against; we therefore compute a pseudo-true value on a single sample of 4×10^5 observations and treat that as the target. For each sample size we draw M independent samples, compute a percentile interval from R bootstrap replicates, and record whether it covers.

Table 4 supports a qualified endorsement. The point estimator is close to unbiased at every sample size and its dispersion falls at the expected rate. Coverage improves with n and is close to nominal by $n = 4000$, but at $n = 500$ the interval under-covers, at 92.0% against a nominal 95%. The direction is what one would expect: the breakdown point is a smooth but nonlinear functional of an estimated variance matrix, and the percentile interval inherits the resulting finite-sample skewness without correcting for it.

The practical recommendation is therefore to treat the interval as approximate in small samples, and to prefer the cluster bootstrap whenever the design has a grouping structure, as in Section 7.8. Bias-corrected or accelerated intervals are a natural improvement and are not currently implemented; we note this as the most useful direction for future work on the package.

n	Coverage (%)	Bias	SD	Mean CI width
500	92.0	-0.0053	0.0294	0.1122
1000	92.0	-0.0030	0.0208	0.0799
4000	95.6	0.0001	0.0097	0.0395

Table 4: Finite-sample performance of the percentile bootstrap interval for the breakdown point, at nominal 95% level. $M = 250$ Monte Carlo replications, $R = 199$ bootstrap draws each, $\bar{c} = 1$. The pseudo-true breakdown point is 0.7170, computed on a single sample of 4×10^5 observations. Bias and SD refer to the point estimator of the breakdown point.

11. Comparison with sensemakr

sensemakr (Cinelli *et al.* 2024) implements Cinelli and Hazlett (2020) and is the tool an R user is most likely to reach for. As Table 1 indicated, the two packages parameterize the same bias decomposition (20) differently, and it is worth seeing the numbers side by side on one specification.

```
R> suppressPackageStartupMessages(library("sensemakr"))
R> fit <- lm(form, data = bfg2020)
R> sm <- sensemakr(model = fit, treatment = "tye_tfe890_500kNI_100_l6",
+                 benchmark_covariates = w1, kd = 1, ky = 1,
+                 q = 1, alpha = 0.05, reduce = TRUE)
R> sm$sensitivity_stats[c("r2yd.x", "rv_q", "rv_qa")]
```

```
   r2yd.x   rv_q  rv_qa
1 0.03931 0.1829 0.1463
```

The robustness value is $RV_{q=1} = 0.1829$: an unobservable explaining more than 18.3% of the residual variation in *both* treatment and outcome would suffice to bring the estimate to zero. The partial R^2 of treatment with outcome is 0.0393. **regsensitivity** reports, for the same specification, a breakdown point of 80.4% at $\bar{c} = 1$.

These two numbers are not comparable, and the temptation to compare them is the main hazard in using both packages. RV_q is a partial R^2, bounded by one by construction and measured against the residual variation in the data. The breakdown point is a ratio of index standard deviations, measured against the explanatory power of the chosen calibration covariates. A specification with weak controls will have a high breakdown point and an unchanged RV_q. The right way to read them jointly is as answers to different questions: RV_q asks how strong an unobservable must be in absolute explanatory terms, and $\bar{r}_X^{\text{bp}}$ asks how strong it must be relative to what we have already adjusted for.

The substantive difference is the treatment of the controls. Because **sensemakr** conditions on exogenous controls, its benchmark bounds are computed by asking how large an unobservable "k times as strong as" a named covariate would be, holding the control structure fixed. As Section 7.7 showed, the calibration covariates in this application have partial R^2 against one another as high as 0.89, so the premise is doubtful and a $\bar{c}$-indexed analysis is the more defensible one. Where the controls are genuinely exogenous the two approaches are comple-

Call	Median (ms)
DMP bounds, $\overline{r}_Y = \infty$, $\overline{c} = 0.1$, 10 grid points	6.1
DMP bounds, $\overline{r}_Y = 2$, $\overline{c} = 1$, 10 grid points (DIRECT-L)	993
DMP breakdown across $\overline{c}$, 11 grid points	5.6
Oster bounds (equality), 21 δ values	6.9
Oster breakdown (sign hypothesis)	5.6
Cluster bootstrap, $R = 99$, BFG application	1,038
Bootstrap (no clustering), $R = 99$, BFG application	595

Table 5: Median timings over five runs on an Apple M4 laptop under R 4.5.2. The DIRECT-L regime is roughly two orders of magnitude slower because it solves a nonconvex global optimization at every grid point; see Section 6.2.

mentary, and we recommend reporting both. A reproducible version of this comparison is bundled at `inst/benchmarks/sensemakr-comparison.R`.

12. Performance

Table 5 reports median single-call timings on the Bazzi *et al.* (2020) application ($n = 2036$, ten calibration controls plus state fixed effects). The pattern follows the regime dispatch of Section 6.2: everything in a closed-form regime costs a few milliseconds, and the DIRECT-L regime costs about a second. The bundled script `inst/benchmarks/performance.R` regenerates every row.

For context, DMP report that the breakdown-frontier computation for their application takes "about 15 seconds on a standard machine." The corresponding **regsensitivity** call is the third row of Table 5. The gain comes from the staging described in Section 6: the variance summary is computed once and reused across every point on the grid, rather than recomputed per evaluation.

13. Summary and discussion

regsensitivity brings the Diegert *et al.* (2026), Oster (2019) and Masten and Poirier (2026) analyses into R as a single coherent package. Its distinguishing feature relative to the existing ecosystem surveyed in Section 5 is that it does not require the included controls to be exogenous, and that it treats the breakdown point as a first-class object which can be traced against a second sensitivity parameter, plotted, and bootstrapped. Section 8 verifies the implementation against every published value of Diegert *et al.* (2026) Table 1 column (5) and Tables 3 and 4, pinned by 26 regression tests so that future changes announce themselves.

Three limitations are worth stating plainly. First, the package implements Assumption A6 only in the one-sided form $[0, \overline{c}]$; DMP's general two-sided restriction $[\underline{c}, \overline{c}]$, which is useful when a researcher wants to assert that controls are *at least* somewhat endogenous, is not available. Second, the region $\overline{r}_X > r_{\max}(\overline{c}) > \overline{r}_Y$ is uncharacterized in the underlying theory and the package raises an error there rather than guessing; Section 7.9 shows when a user meets this boundary. Third, the bootstrap interval is a percentile interval and, as Section 10 documents, under-covers at $n = 500$. Adding bias-corrected and accelerated intervals is the highest-value extension we can identify, and the simulation harness in Section 10 provides the yardstick against which to judge it.

The package, with its five vignettes—a tour, a paper-replication walk-through, a stylized illustration of the Masten–Poirier extensions, a plotting-and-tables guide, and a Stata migration guide—is available at `https://github.com/mikenguyen13/regsensitivity`.

Computational details

The results in this paper were obtained using R 4.5.2 on aarch64-apple-darwin20, with the packages **regsensitivity** 0.1.2, **ggplot2** 4.0.0, **nloptr** 2.2.1, **AER** 1.2.15, **sensemakr** 0.1.6 and **knitr** 1.50. R itself and all packages used are available from the Comprehensive R Archive Network (CRAN) at `https://CRAN.R-project.org/`. Every number, table and figure in this paper is regenerated by the replication script `paper.R` distributed with the manuscript; the two computationally expensive chunks—the breakdown frontier of Section 7.9 and the coverage study of Section 10—are cached, and the full build takes a few minutes from a cold cache.

Acknowledgments

This R package implements the methodology of Diegert *et al.* (2026), Oster (2019) and Masten and Poirier (2026). The package internals follow the structure of the Stata code released by Diegert, Masten and Poirier alongside their paper, and several internal function names are retained from it to make the two implementations comparable.

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A. From the identified set to the breakdown point

The closed form for the breakdown point in (15) follows from Theorem 1 by a short argument that is worth setting out, because it makes clear why $\bar{r}_X^{\text{bp}} < 1$ is a property of the arbitrarily-endogenous-control case rather than a general fact.

Take $\beta_{\text{med}} > 0$ and the hypothesis $\beta_{\text{long}} > 0$. By (12) the identified set is the interval $\beta_{\text{med}} \pm \text{dev}(\bar{r}_X)$, so the hypothesis holds for every element of the set exactly when $\beta_{\text{med}} - \text{dev}(\bar{r}_X) \geq 0$. Since dev is continuous and strictly increasing in $\bar{r}_X$ on the region where the bounds are finite, and $\text{dev}(0) = 0$, the supremum in (14) is attained where the lower bound just touches zero:

$$\text{dev}(\bar{r}_X^{\text{bp}}) = \beta_{\text{med}}. \tag{23}$$

Substituting (13) and writing $\lambda := R^2_{X\sim W_1}$ and $V := \text{var}(Y^{\perp X, W_1}) / \text{var}(X^{\perp W_1})$, condition (23) is

$$\frac{\bar{r}_X^2 \lambda}{1 - \lambda - \bar{r}_X^2 \lambda} \cdot V = \beta^2_{\text{med}}.$$

Solving for $\bar{r}_X^2$ gives

$$\bar{r}_X^2 = \frac{\beta^2_{\text{med}}(1-\lambda)}{\lambda\,(V + \beta^2_{\text{med}})},$$

and rewriting $\beta^2_{\text{med}}/(V + \beta^2_{\text{med}})$ as the partial R^2 of the outcome on treatment, $R^2_{Y\sim X \cdot W_1}$, yields (15). The ceiling is now visible: the right-hand side of (15) is a ratio whose denominator exceeds its numerator by $(1-\lambda)/\lambda > 0$, so $\bar{r}_X^{\text{bp}} < 1$ whenever $\lambda < 1$. Under Theorem 3 the deviation function is driven by $\bar{z}_X(\bar{r}_X, \underline{c}, \bar{c})$ instead, and restricting $\bar{c}$ below one slows its growth in $\bar{r}_X$, which is what allows the breakdown point to exceed one—as it does at $\bar{c} \in \{0, 0.1\}$ in Section 7.1.

The same argument run against a general hypothesis value b rather than zero replaces β_{med} by $|\beta_{\text{med}} - b|$ throughout, which is why the breakdown points reported in Section 7 decline as b moves toward β_{med}.

B. Reproducing the results in this paper

Every number, table and figure above is produced by the replication script `paper.R` distributed with this manuscript, which runs top to bottom in a clean session. Its structure follows the sections of the paper. The two expensive steps are the breakdown frontier of Section 7.9 and the coverage study of Section 10; in the manuscript these are **knitr**-cached, and in the standalone script they are plain code with their run times noted in comments.

The regression tests that pin the replicated values are run with

```
R> testthat::test_local()
```

from the package source, or on the installed package with

```
R> testthat::test_package("regsensitivity")
```

The benchmark timings of Table 5 and the **sensemakr** comparison of Section 11 are regenerated by

```
R> source(system.file("benchmarks", "performance.R",
+                     package = "regsensitivity"))
R> source(system.file("benchmarks", "sensemakr-comparison.R",
+                     package = "regsensitivity"))
```

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